

Music Genre Classification (Project Mentor: Akanksha)

Background

Music Genre Recognition (MGR) is important for recommendation systems, playlist creation, and personalized streaming. Manual categorization is difficult and inconsistent. Machine Learning and audio signal processing can automatically analyze acoustic features (e.g., spectral features and MFCCs) to assign a genre label to a song.

Problem Statement

The objective of this project is to design a machine learning system that automatically classifies an input audio track into its correct music genre. Given a short music clip, the model should extract meaningful audio features and predict the most probable genre (e.g., classical, jazz, pop, rock, etc.) with high accuracy.

Dataset

Dataset Name: GTZAN Dataset – Music Genre Classification

(<https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification>)

The dataset contains **raw audio files, handcrafted feature CSV files, and spectrogram images** for each music track. For handcrafted features, two CSV files are provided:

- **Audio-level features (30-second clip):** one row per song containing mean and variance statistics of extracted audio features (e.g., chroma, MFCC, spectral features).
- **Chunk-level features (3-second clips):** each song is divided into ten 3-second segments, increasing the number of samples by 10×.

NOTE: For all tasks below, maintain a **consistent song-level train–test split** across every method to enable a fair and comparative evaluation.

TASK 1. Handcrafted Feature-based Classification (1+10+2+2=15 Marks)

- A. Visualize the dataset in tabular form to understand and report the feature names, number of samples, feature dimensions, class distribution.
- B. Train **two machine learning models of your choice** (e.g., Logistic Regression, SVM, Random Forest, Gradient Boosting) on:
 - a. (i) **30-second feature CSV** (one sample per song)

- b. (ii) **3-second feature CSV** (multiple samples per song). For the 3-second dataset, aggregate segment predictions to obtain a final song-level label using a voting strategy (e.g., majority voting or probability averaging).
- C. For both datasets, report the Classification Accuracy, Macro F1-score and Confusion Matrix.
- D. Compare the performances and comment on:
 - a. Effect of model selection.
 - b. Effect of dataset size (fewer long samples vs many short samples)

TASK 2. Dimensionality Reduction using PCA (4+5+1= 10 Marks)

- A. Select the **best-performing model from Q1** and apply **Principal Component Analysis (PCA)** to the feature set prior to training.
- B. Re-train the model and report the evaluation metrics.
- C. Discuss whether dimensionality reduction impacts the performance.

TASK 3. Spectrogram-based Deep Learning (8+2= 10 Marks)

Using the provided spectrogram images:

- A. Design and train a **2D Convolutional Neural Network (CNN)** for music genre classification. Choose appropriate hyperparameters (learning rate, batch size, number of layers, etc.).
- B. Report the main hyperparameters chosen and the evaluation metrics.

TASK 4. Pretrained Audio Representation + Classifier (8+2= 10 Marks)

- A. Pretrained audio representation models (e.g., music-audio embedding models such as MERT, or similar music-focused representation learners) can learn rich acoustic features from raw audio.
 - a. Use the **raw waveform or spectrogram** as input to a pretrained music-audio representation model.
 - b. Extract embeddings for each track.
 - c. Train a **small MLP classifier (2–3 fully connected layers)** on top of these embeddings to predict the music genre. Feel free to design and use an architecture of your choice for the downstream classifier.
- B. Report the Classification metrics & briefly discuss how pretrained representations help downstream tasks such as music genre recognition.

TASK 5. Comparative Analysis (5 Marks)

Compare the performance of all approaches in the above questions. Analyze and explain:

- Which method performs best and why?
- Trade-offs between traditional ML and deep learning approaches.

- Impact of data representation (features vs images vs embeddings) on classification performance.